



Activity File: Permissions

During the last activity, we cleaned up the system's groups and users, and removed all the malicious users. We also removed users from the `sudo` group, and found and removed a rogue group.

Your senior administrator now wants you to secure a few important files and directories. Recall that during our scavenger hunt yesterday, we talked about the sensitive files on the system that contain all the users (`/etc/passwd`), the passwords (`/etc/shadow`), and the groups (`/etc/group`).

This activity will give you an opportunity to practice inspecting and setting file permissions on these sensitive files. Use the checklist provided by your senior administrator in the Instructions section for information on which files to inspect and modify permissions for.

You'll use the same lab environment you used in the previous exercises:

- Username: `sysadmin`
- Password: `cybersecurity`

Instructions

Your senior administrator has asked you to complete the following:

1. Set permissions on `/etc/shadow` to allow only `root` read and write access.
2. Set permissions on `/etc/gshadow` to allow only `root` read and write access.
3. Set permissions on `/etc/group` to allow `root` read and write access, and all others read access only.
4. Set permissions on `/etc/passwd` to allow `root` read and write access, and all others read access only.

Bonus

5. Verify all accounts have passwords.
6. Recall that if any user has the UID of `0`, the system thinks they are `root`. Verify that no users have a UID of `0` besides `root`. If you find one that does, change the user's UID to any value greater than `1000`.
7. Provide a list of all permission changes that you make in a text file in your research directory.

Attempt to complete these tasks on your own. If you need help, refer to the further instructions below.

Further Instructions

- Start by inspecting the file permissions on each of the files listed. Determine if they are already set correctly or if you need to change the permissions.
- Use [this web resource](#) if you get stuck.
- Verify that all accounts have passwords.
 - If you get stuck, google how to determine if an account has a password on a Linux system.
- Verify that only the root user has a UID of 0.
Hint: This is similar to verifying the password.
- Document all of your findings into a file using nano. Keep that file in your research directory.